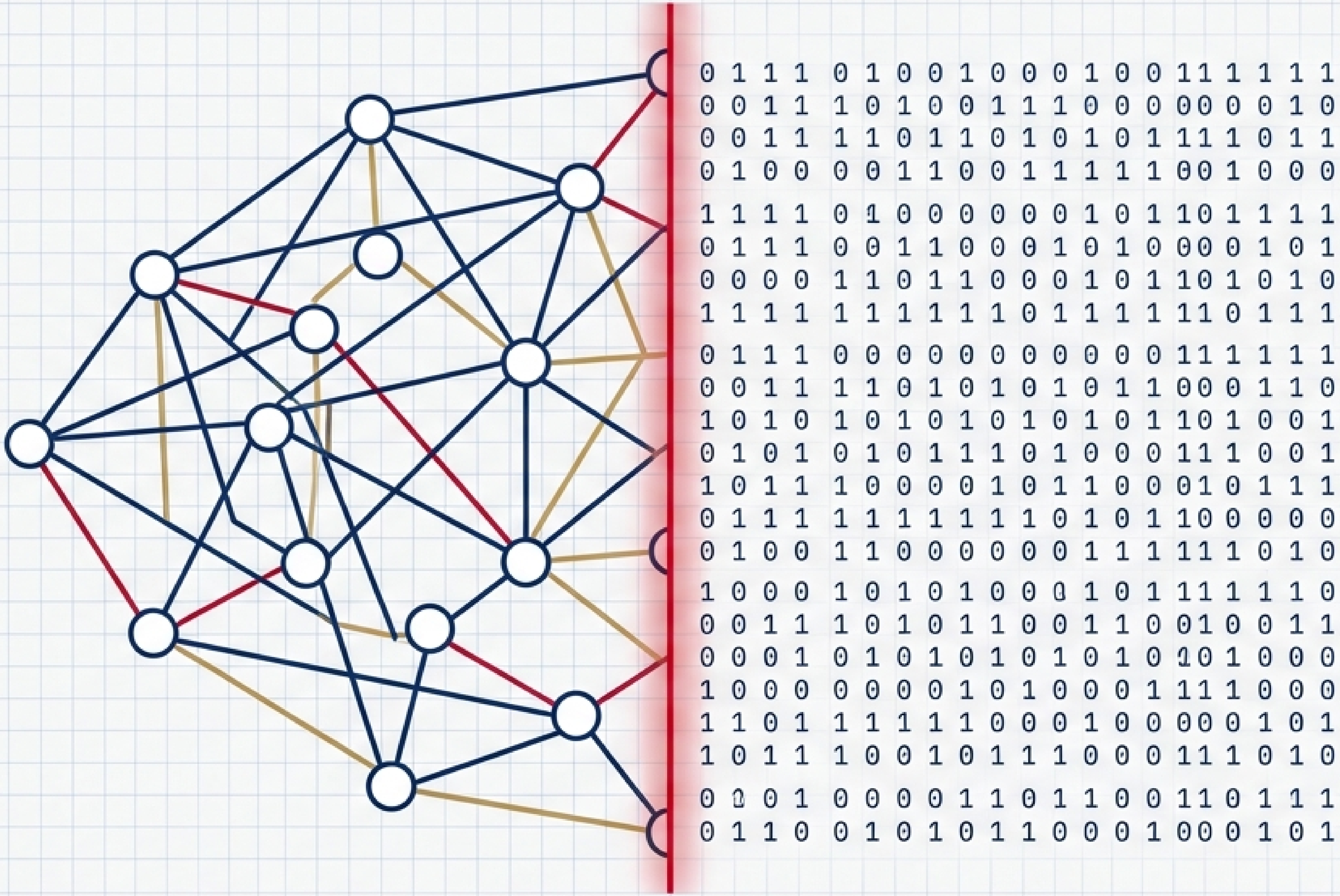
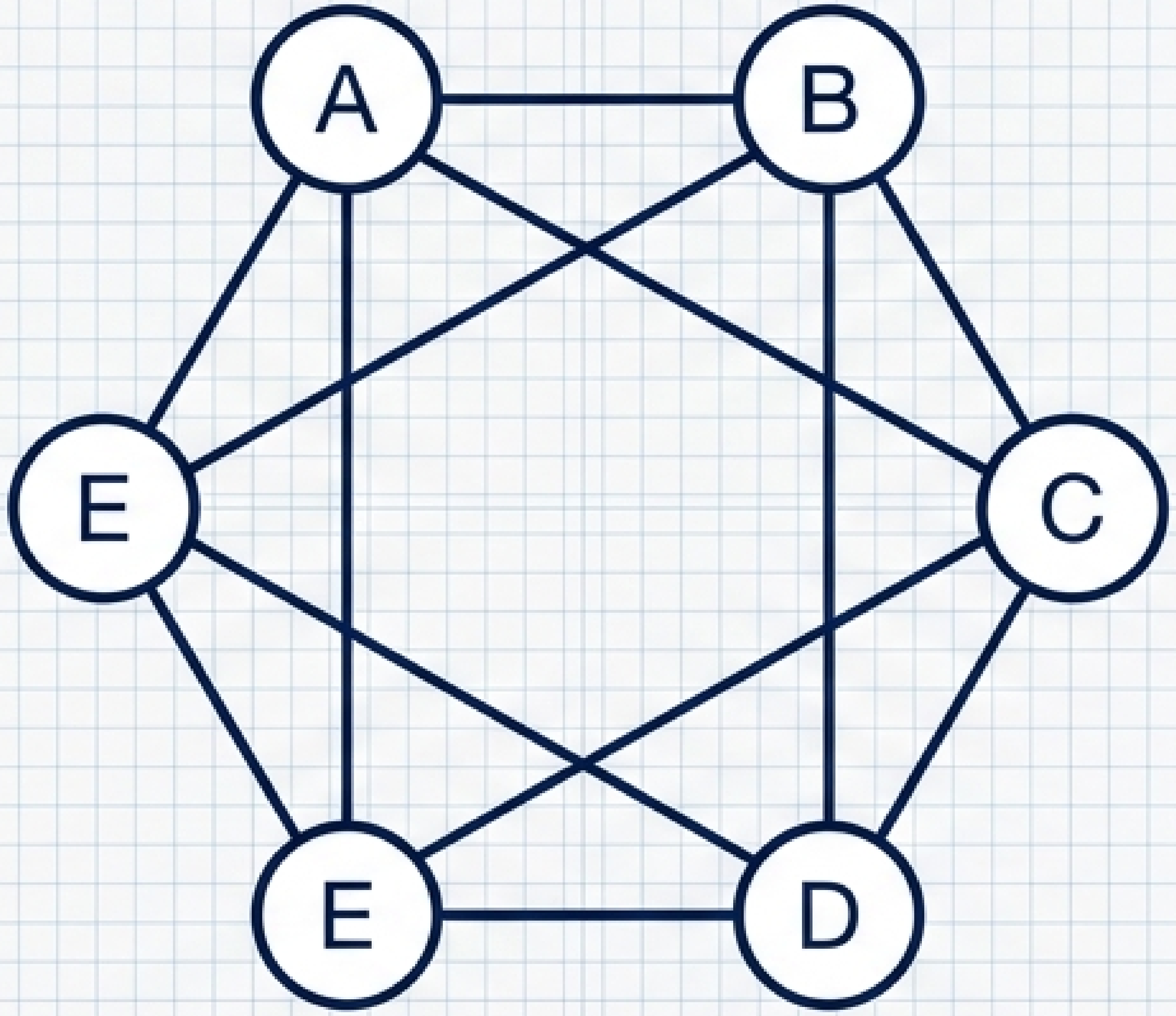


Decoding the Invisible Geometry

A Visual Primer on Graph Theory, Matrices, and Algorithmic Optimization

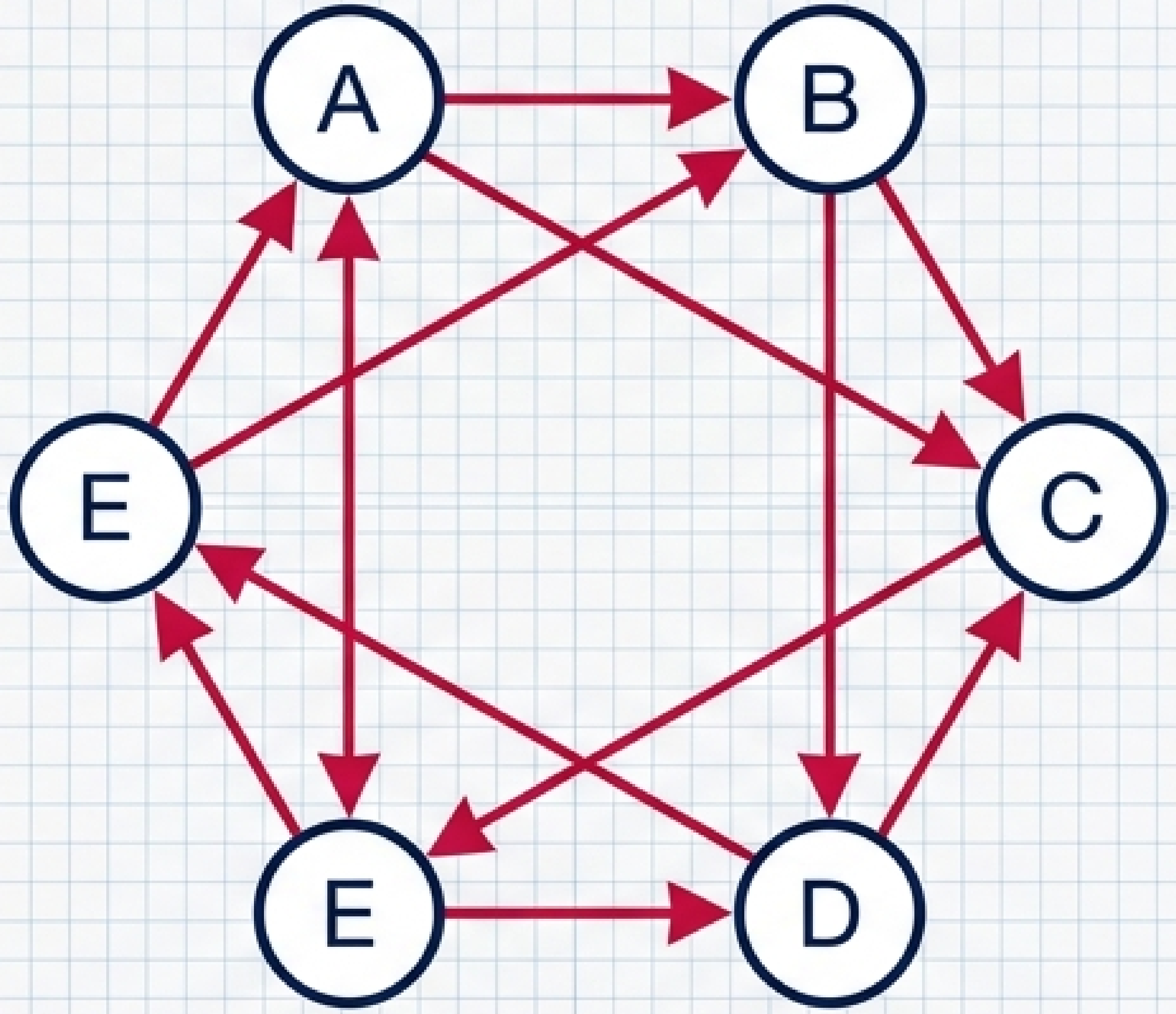
Designed for Engineering & Computer Science | Based on BSC-301 Curricula

The Undirected Graph

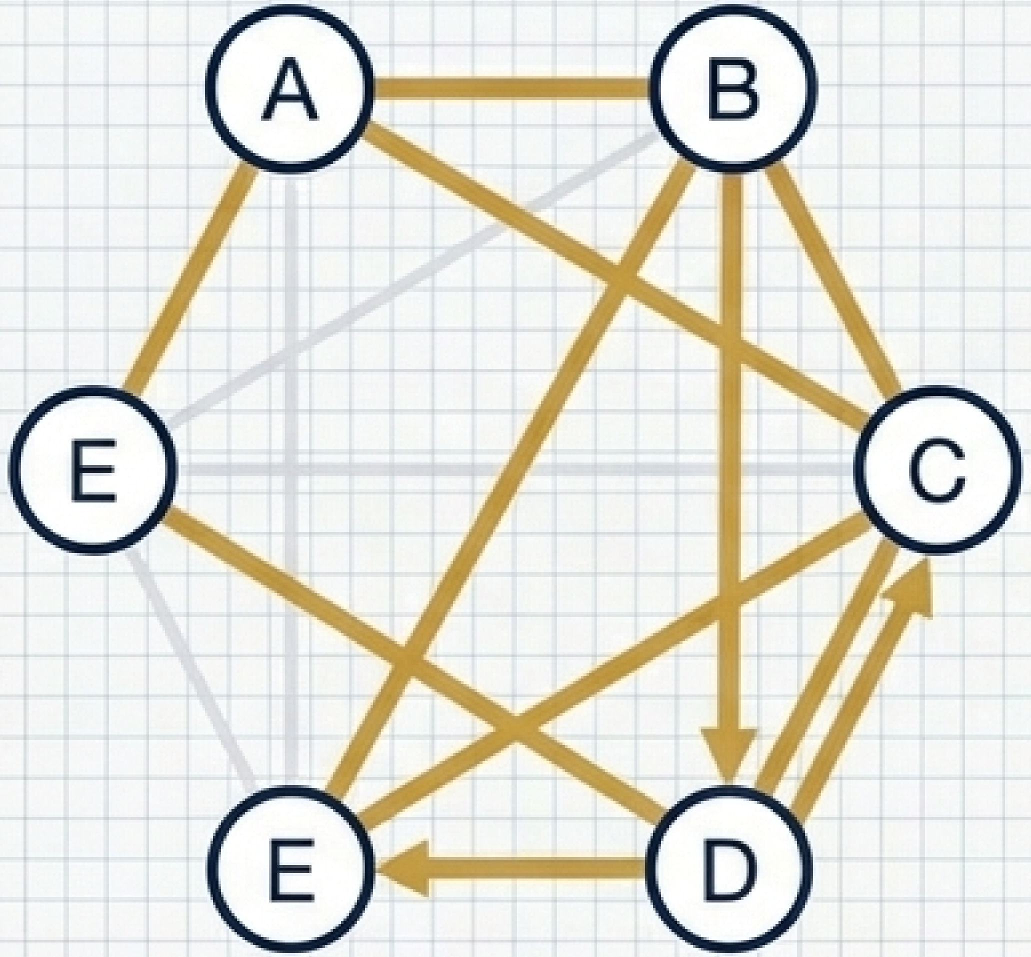


Graph: A set of Vertices (nodes) connected by Edges (links). Movement is unrestricted and bi-directional.

The Digraph

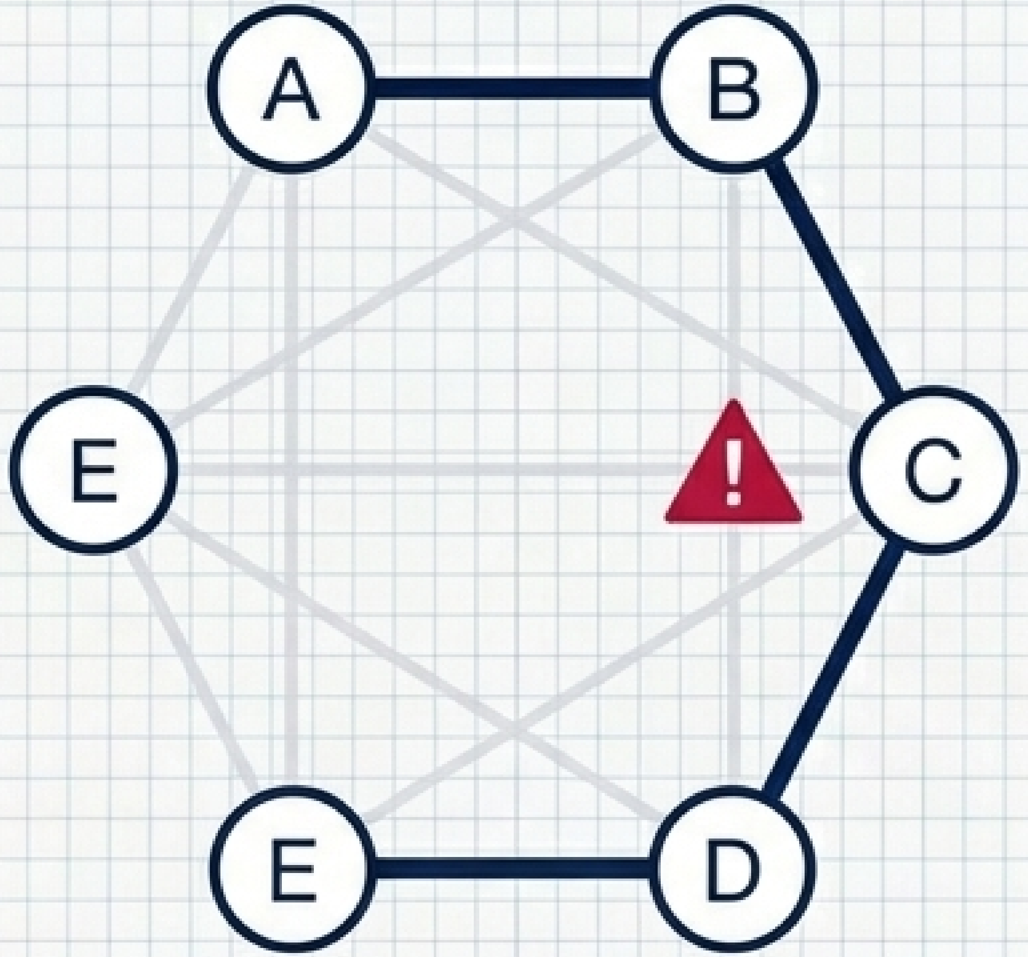


Digraph (Directed Graph): Edges have a strict direction. The path from A to B does not guarantee a path from B to A.



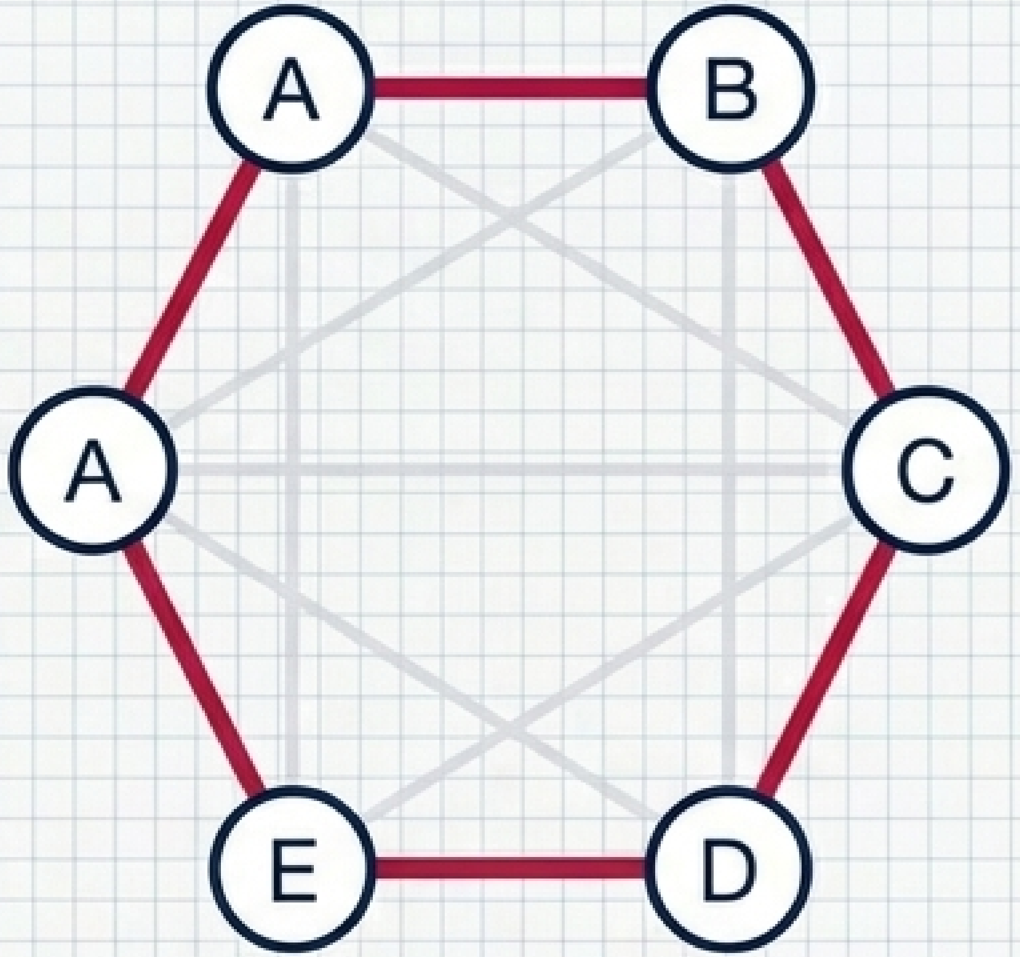
Walk

Free traversal. Vertices and edges can repeat.



Path

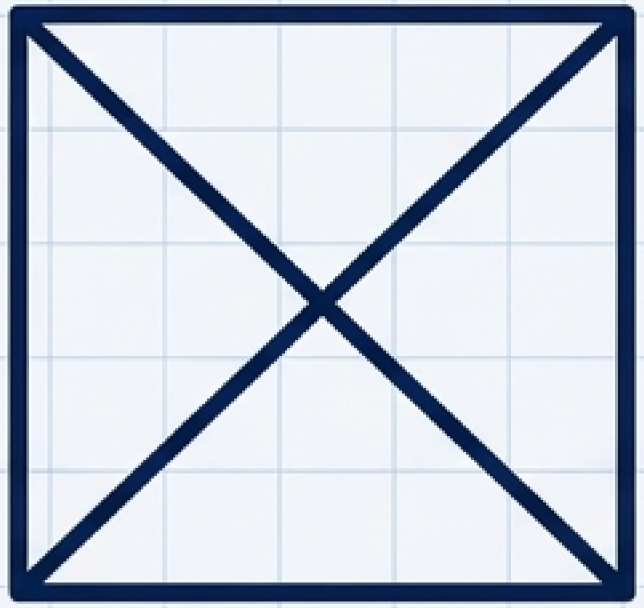
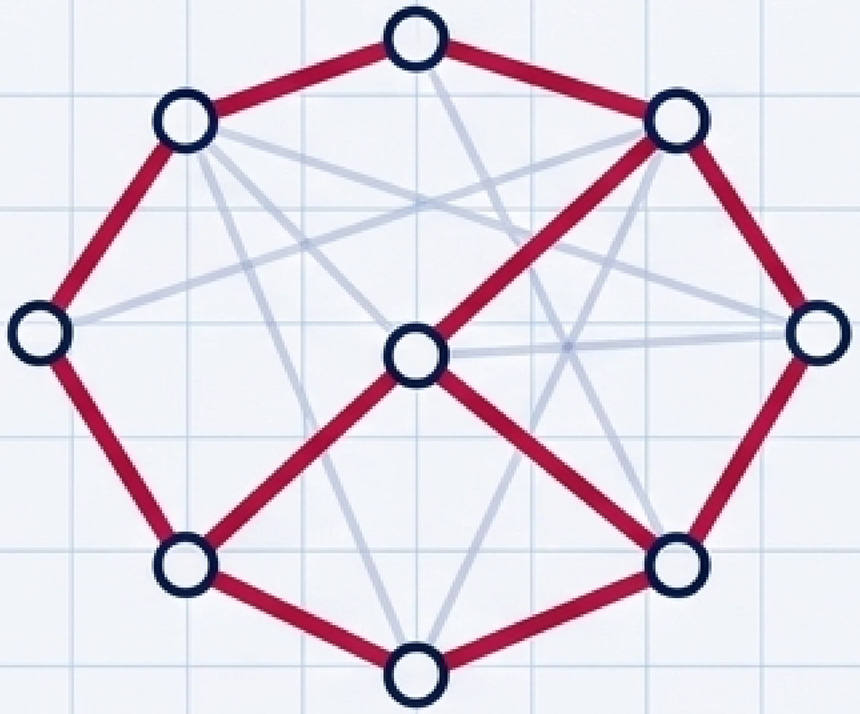
Strict traversal. No vertices or edges can repeat.



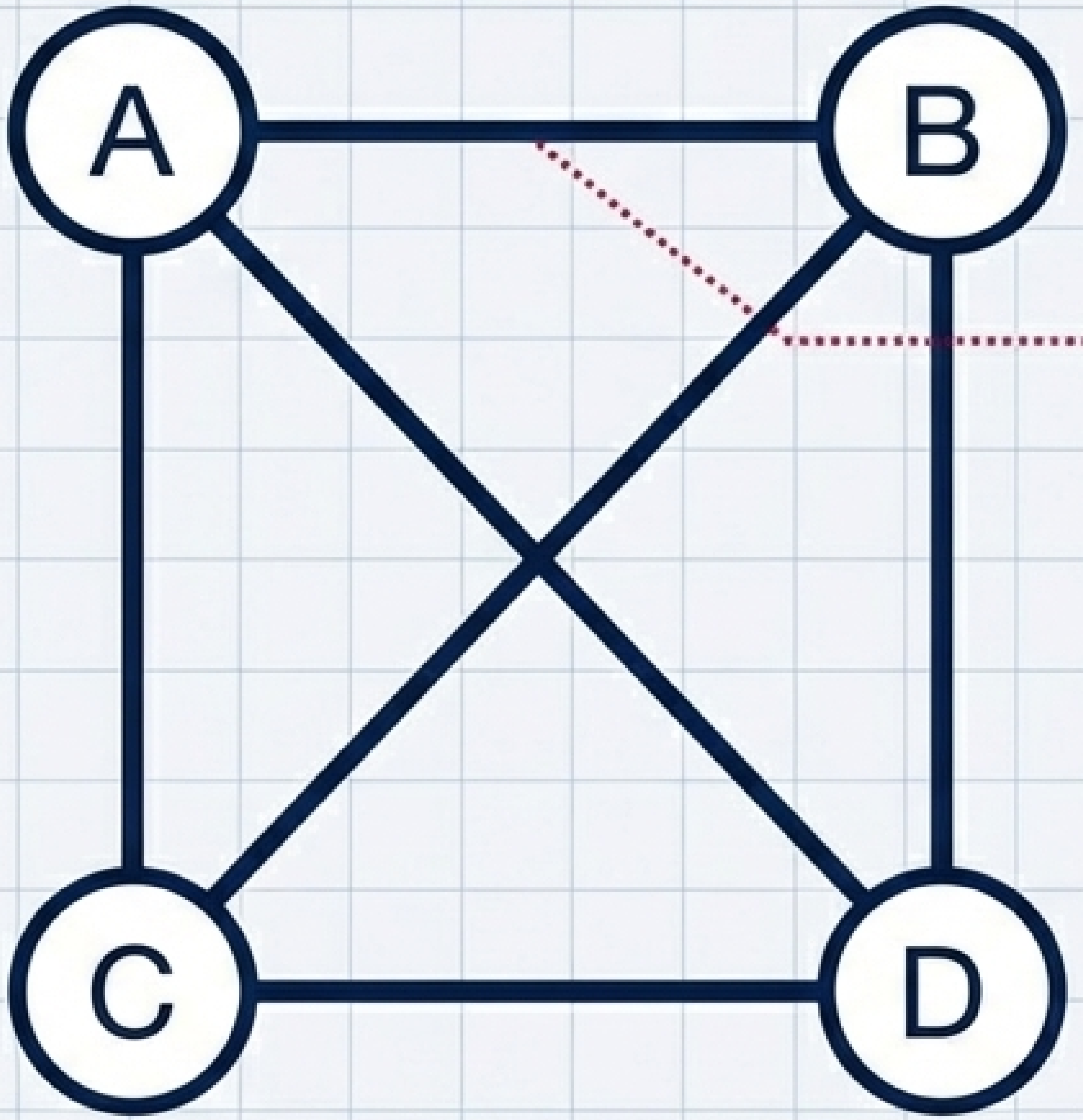
Circuit

Closed loop. Starts and ends on the same vertex. Edges cannot repeat.

The Routing Matrix: Imposing Rules on Chaos

		Eulerian	Hamiltonian
1	The Core Target	The Edges (Links)	The Vertices (Nodes)
2	The Golden Rule	Every single edge must be traversed exactly once.	Every single vertex must be visited exactly once.
3	Visual Example		

Adjacency Matrix



	A	B	C	D
A	0	1	1	1
B	1	0	1	1
C	1	1	0	1
D	1	1	1	0

Computers cannot “see” shapes. Matrices translate geometric relationships into the strict rows and columns required for algorithmic processing.

The Translation Matrix: Data Structures

Adjacency Matrix

Dimension: V x V (Vertices by Vertices).

Logic: A 1 indicates an edge exists between two vertices.

Best For: Dense graphs. Rapidly checking if two specific nodes are connected.

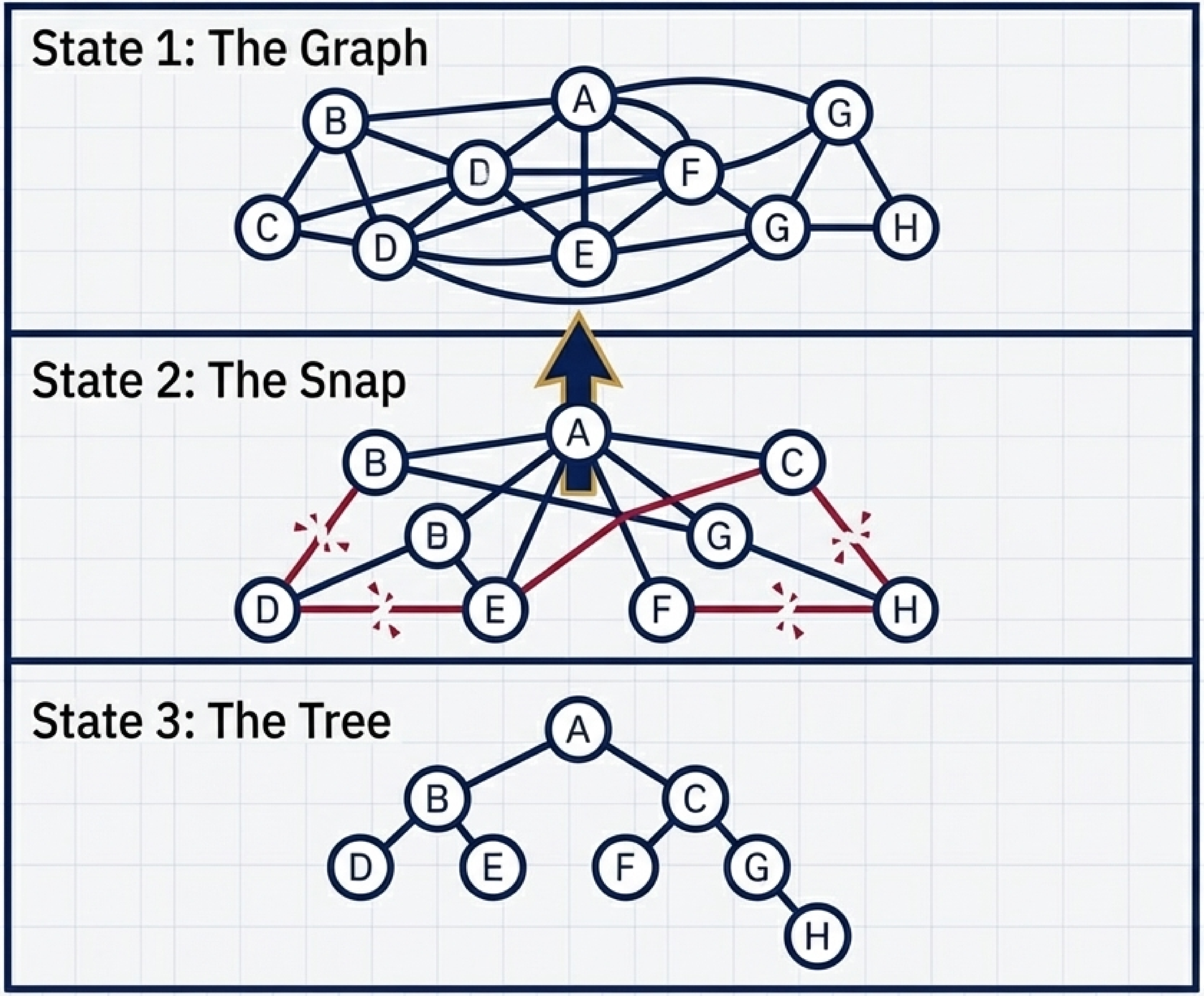
Incidence Matrix

Dimension: V x E (Vertices by Edges).

Logic: A 1 indicates a vertex is connected to a specific edge.

Best For: Sparse graphs. Analyzing specific edge relationships.

If a graph has 5 vertices and 7 edges then write the size of its adjacency matrix.	Answer: 5 x 5 (It depends strictly on Vertices: V x V).

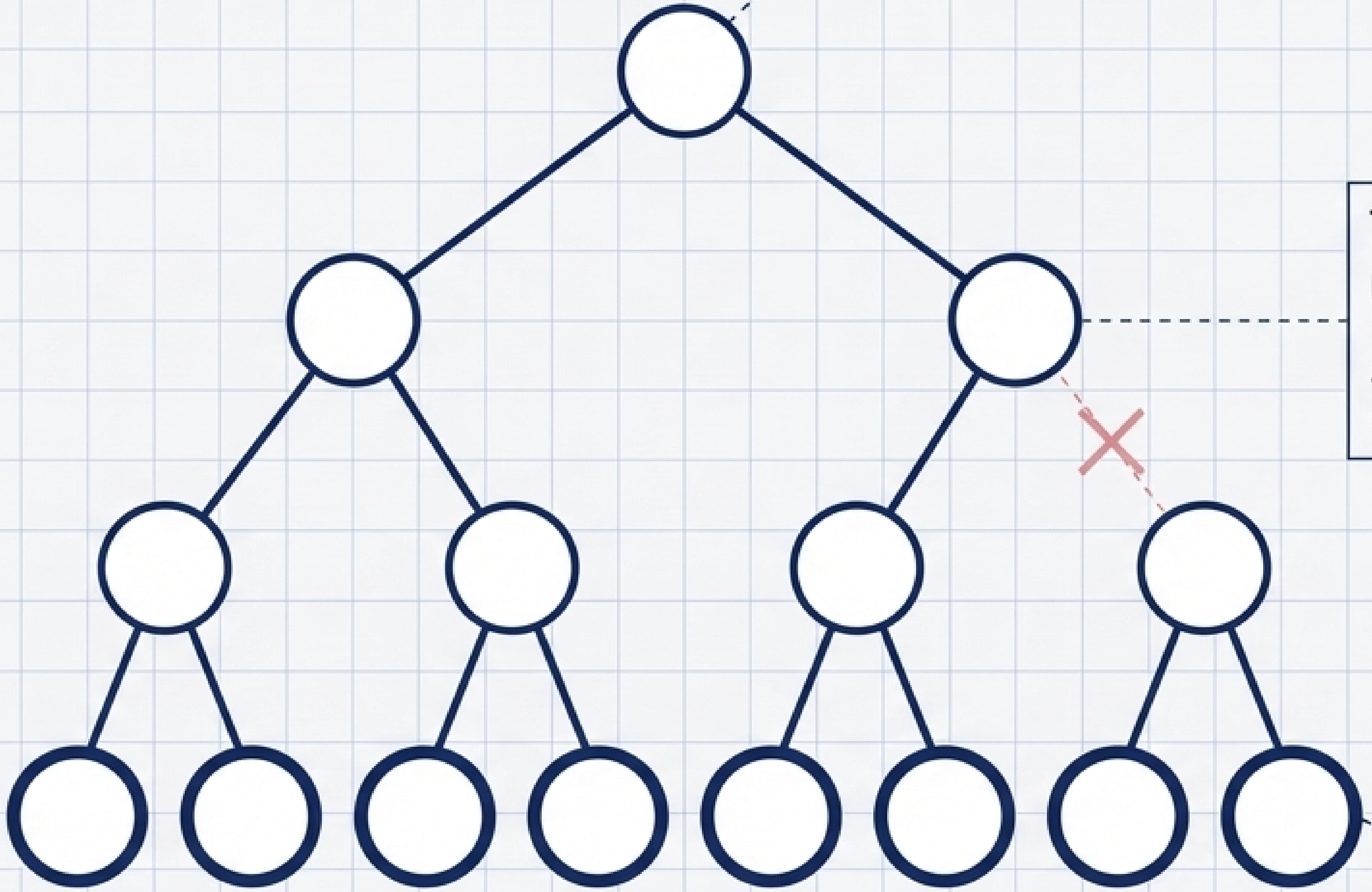


Tree: A connected, undirected graph with absolute zero circuits. It is the ultimate expression of hierarchical order.

Fast Fact: A tree with N vertices always contains exactly N-1 edges.

Binary Tree

A highly structured tree where every node is restricted to a maximum of two descendants.



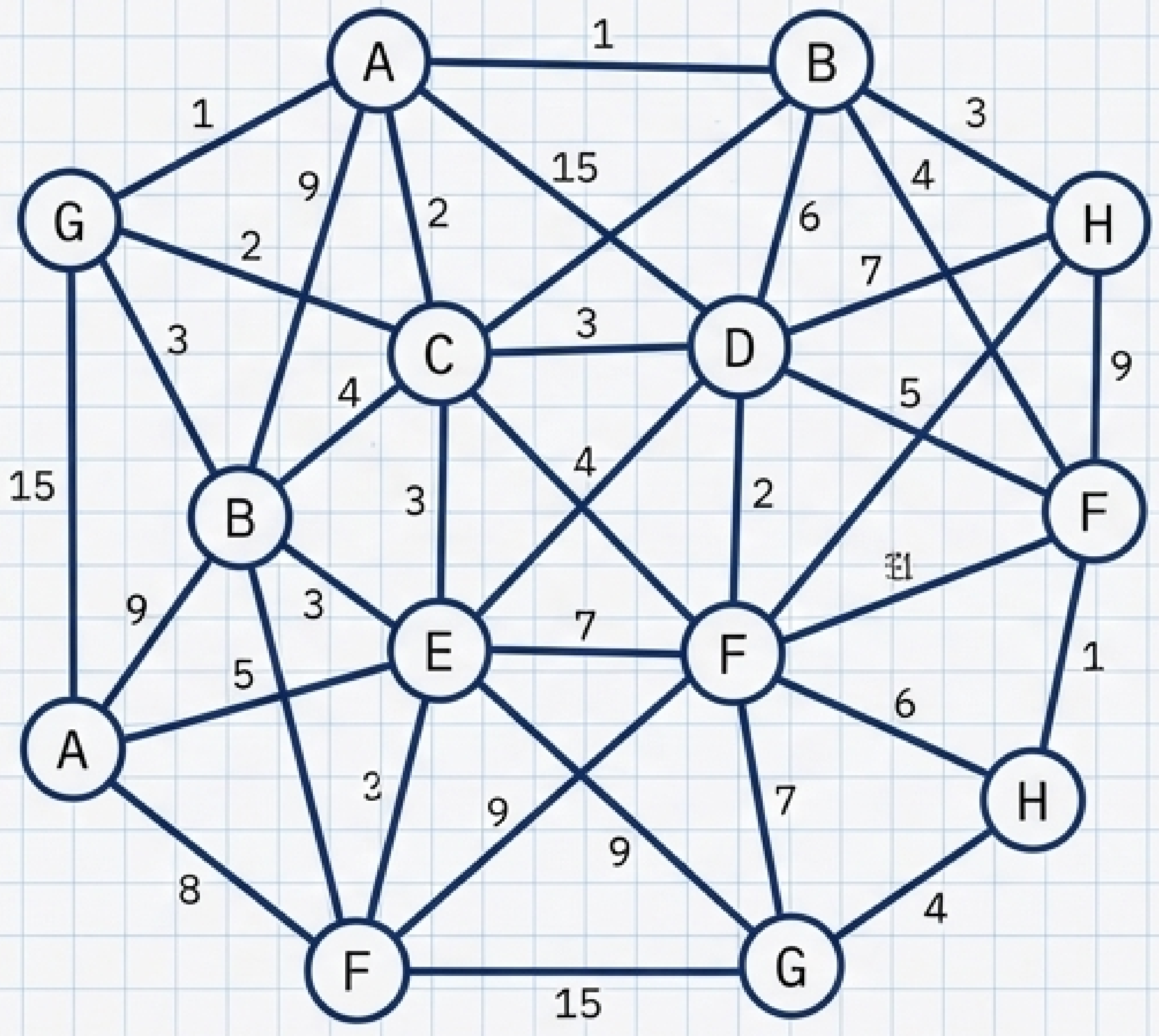
The Root
The singular top-level vertex.

The Constraint
Degree <= 2. Every node has exactly a Left Child and a Right Child. A third branch is impossible.

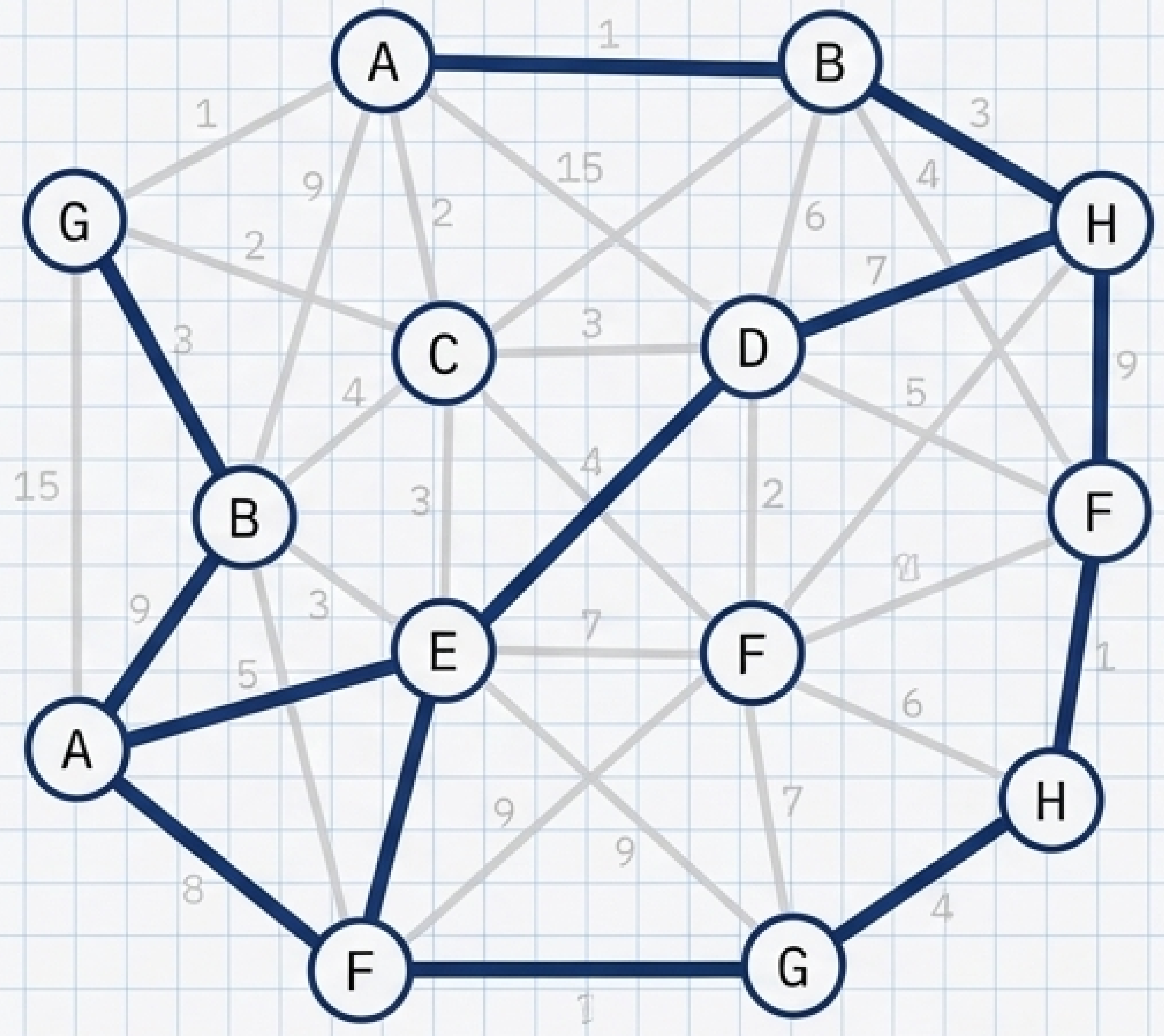
The Leaves
The bottom-tier vertices with zero children.

Minimal Spanning Tree (MST): Before & After

Before: Complex Network with Weighted Edges



After: Minimal Spanning Tree (MST) Subgraph



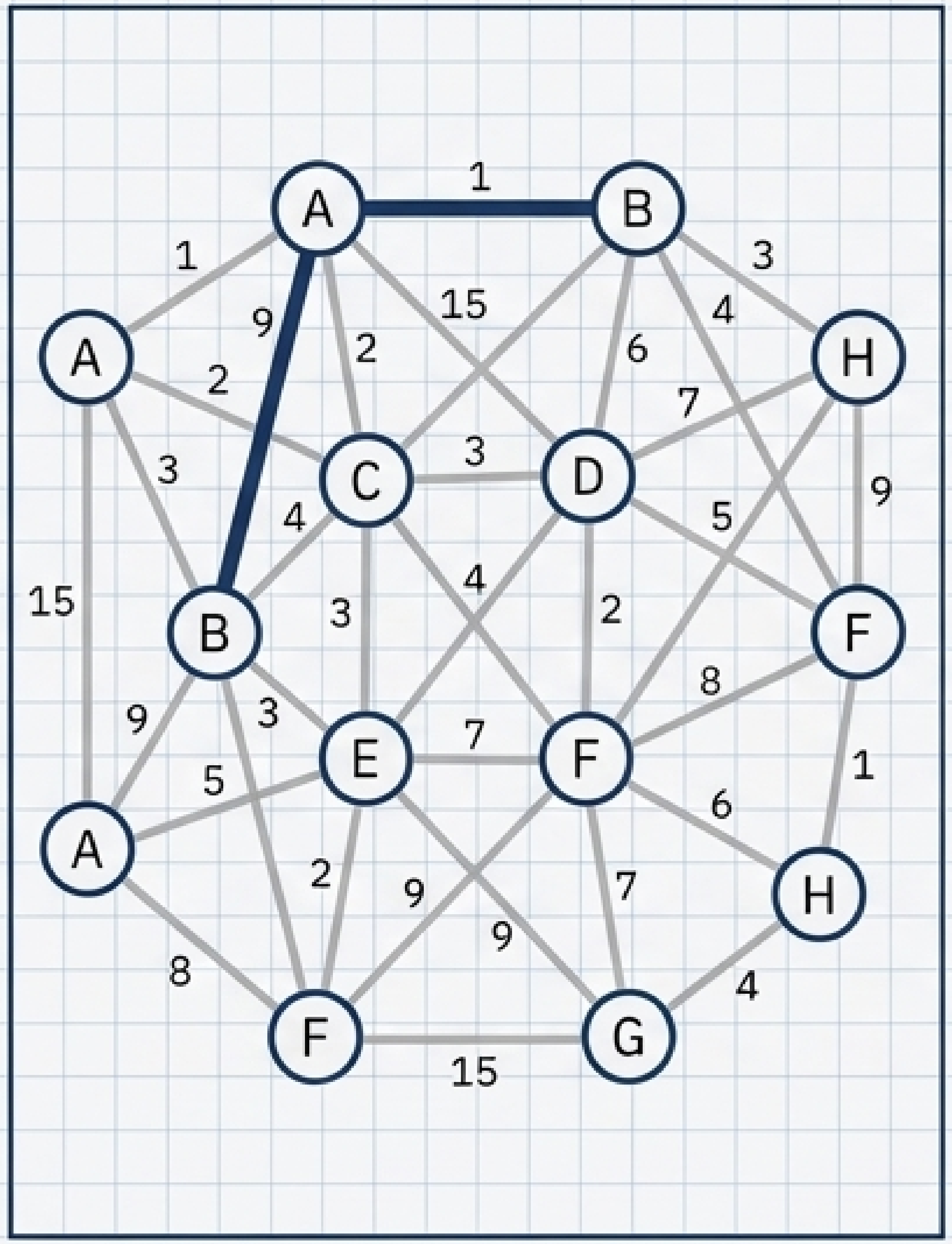
Spanning Tree: A subgraph that connects all vertices of the original graph but strips away enough edges to become a tree (no circuits). The ultimate goal is the **Minimal Spanning Tree (MST):** finding the single spanning tree with the lowest possible total edge weight.

The Optimization Matrix: Solving the MST

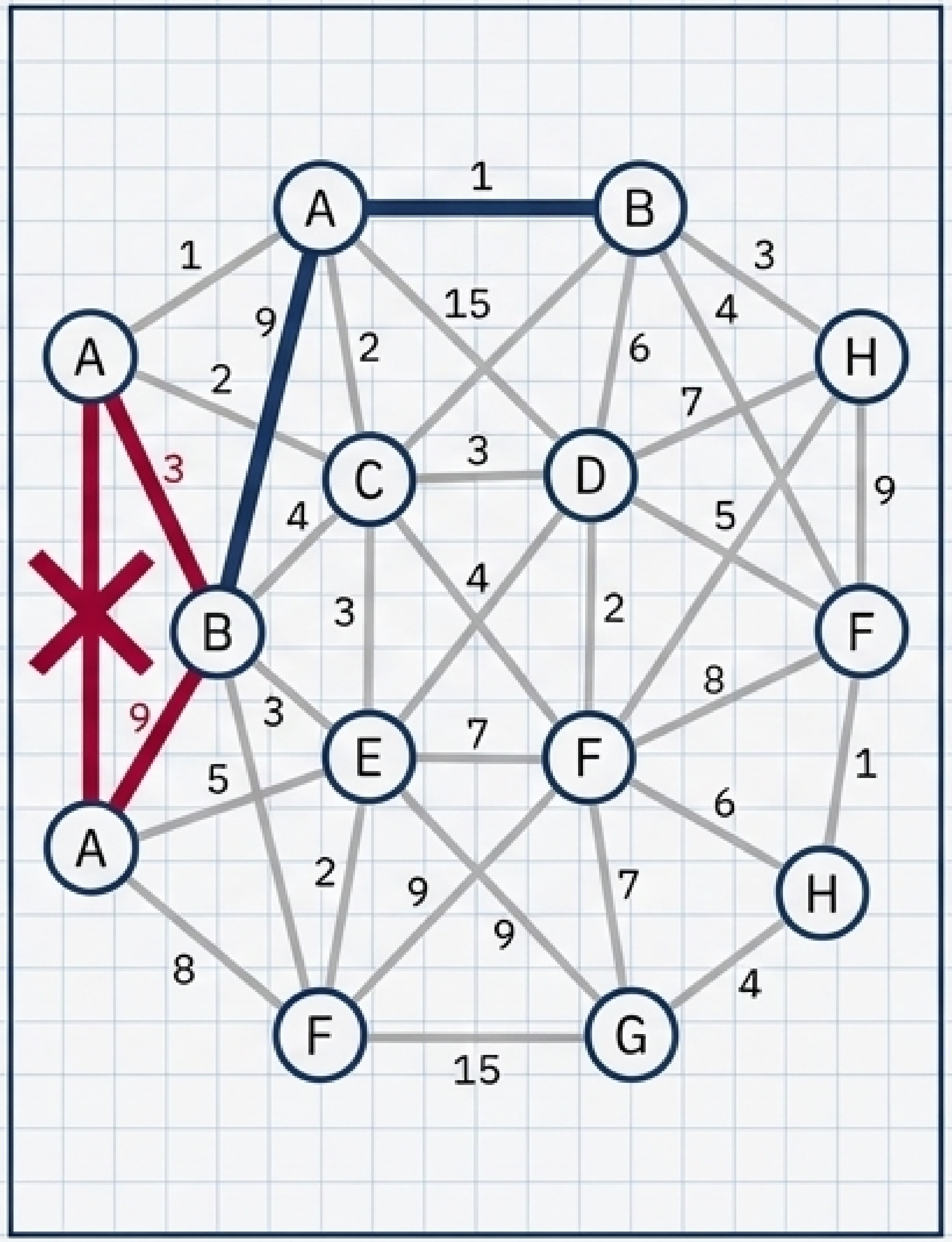
Kruskal's Algorithm (The Edge Sorter)	Prim's Algorithm (The Vertex Expander)
	
Philosophy: Global approach. Sort all edges by weight, pick the smallest, build a forest until it connects into a tree.	**Philosophy:** Local approach. Start at one node, evaluate all immediate neighbors, absorb the closest one into a growing cloud.
Data Focus: Edge-centric.	**Data Focus:** Vertex-centric.

Kruskal's Algorithm: Sort & Connect

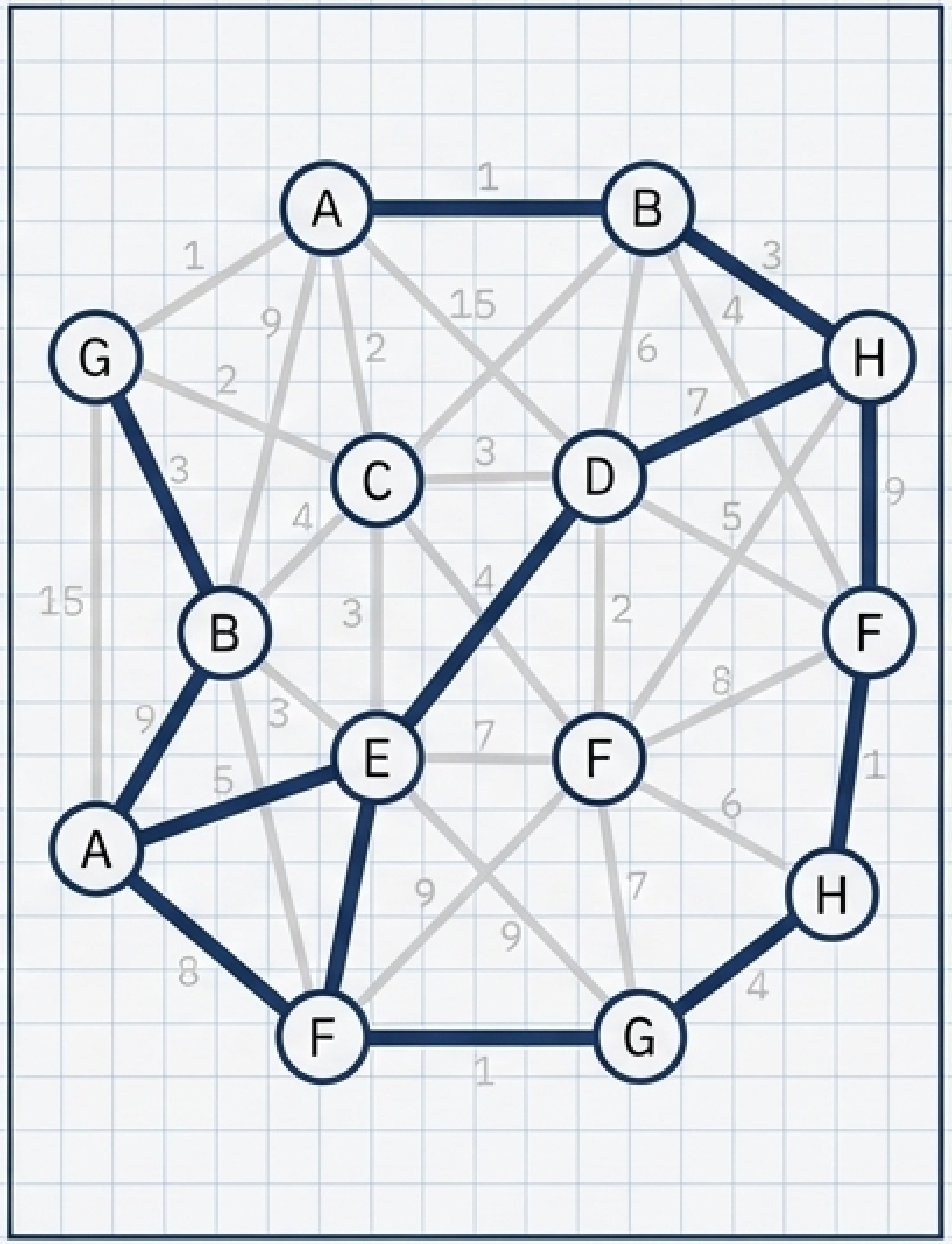
= Unexplored
= Added to MST
= Rejected (Causes Cycle)



Select globally lightest edges.



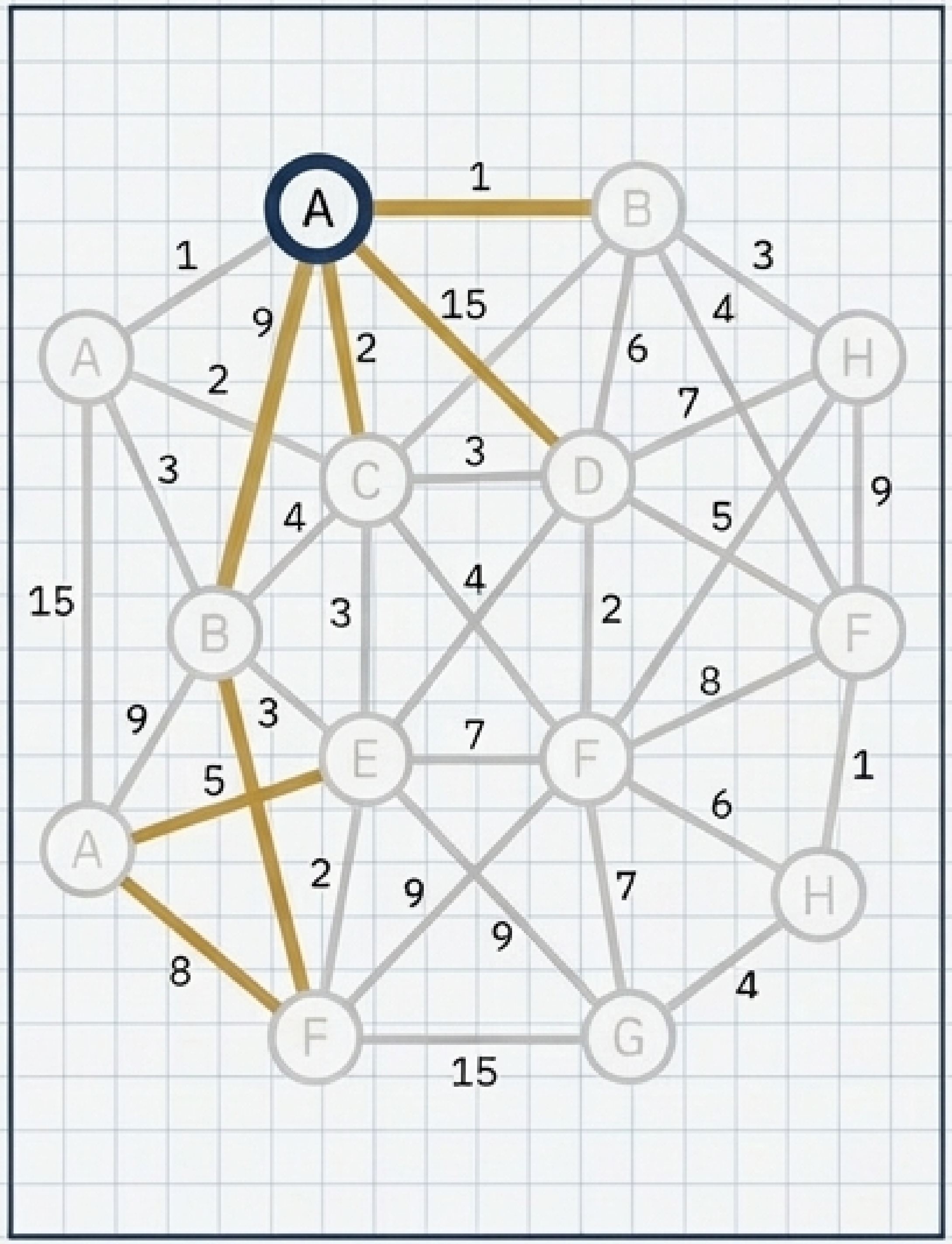
Discard edges that form circuits.



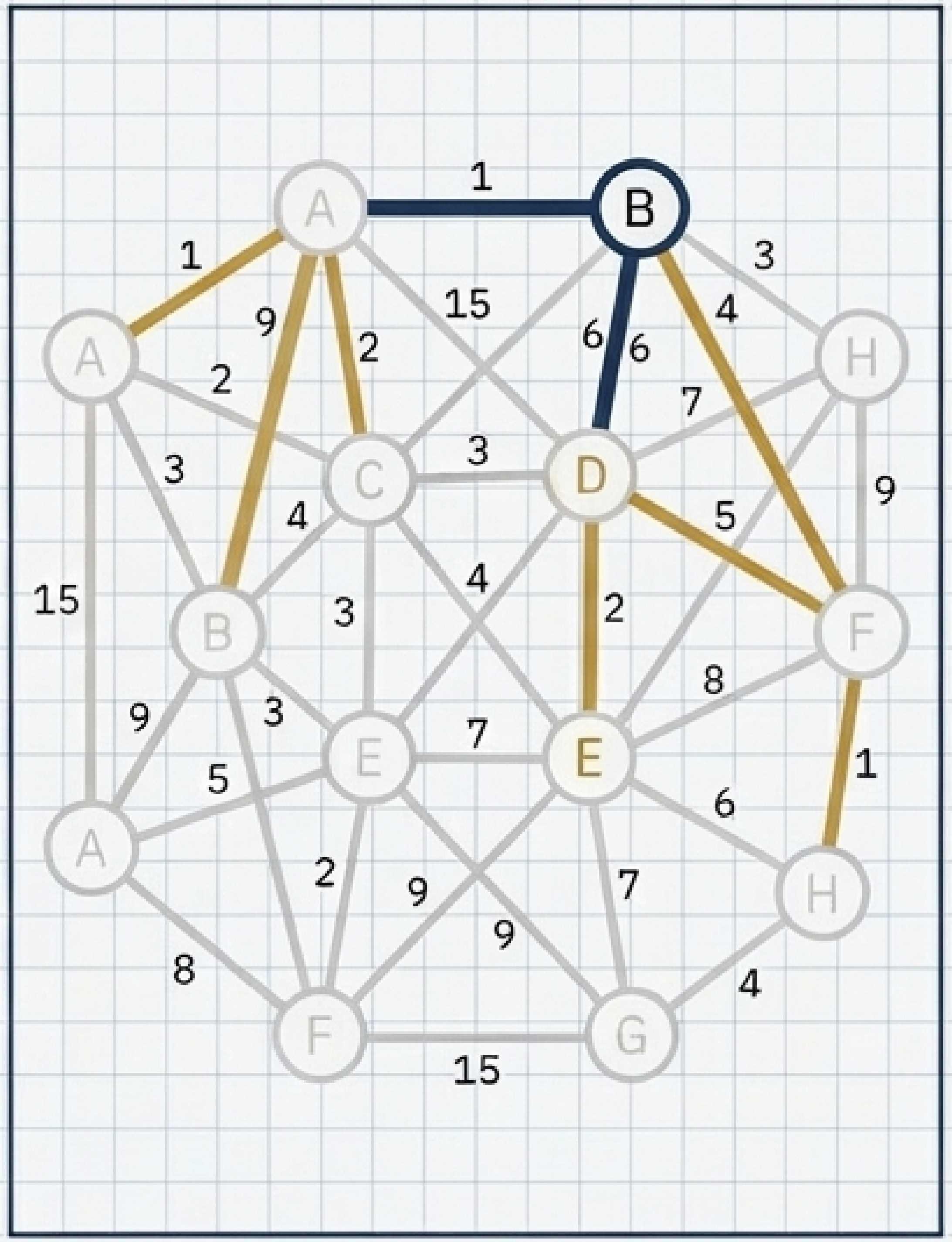
Stop when V-1 edges are added.

Prim's Algorithm: Expand & Absorb

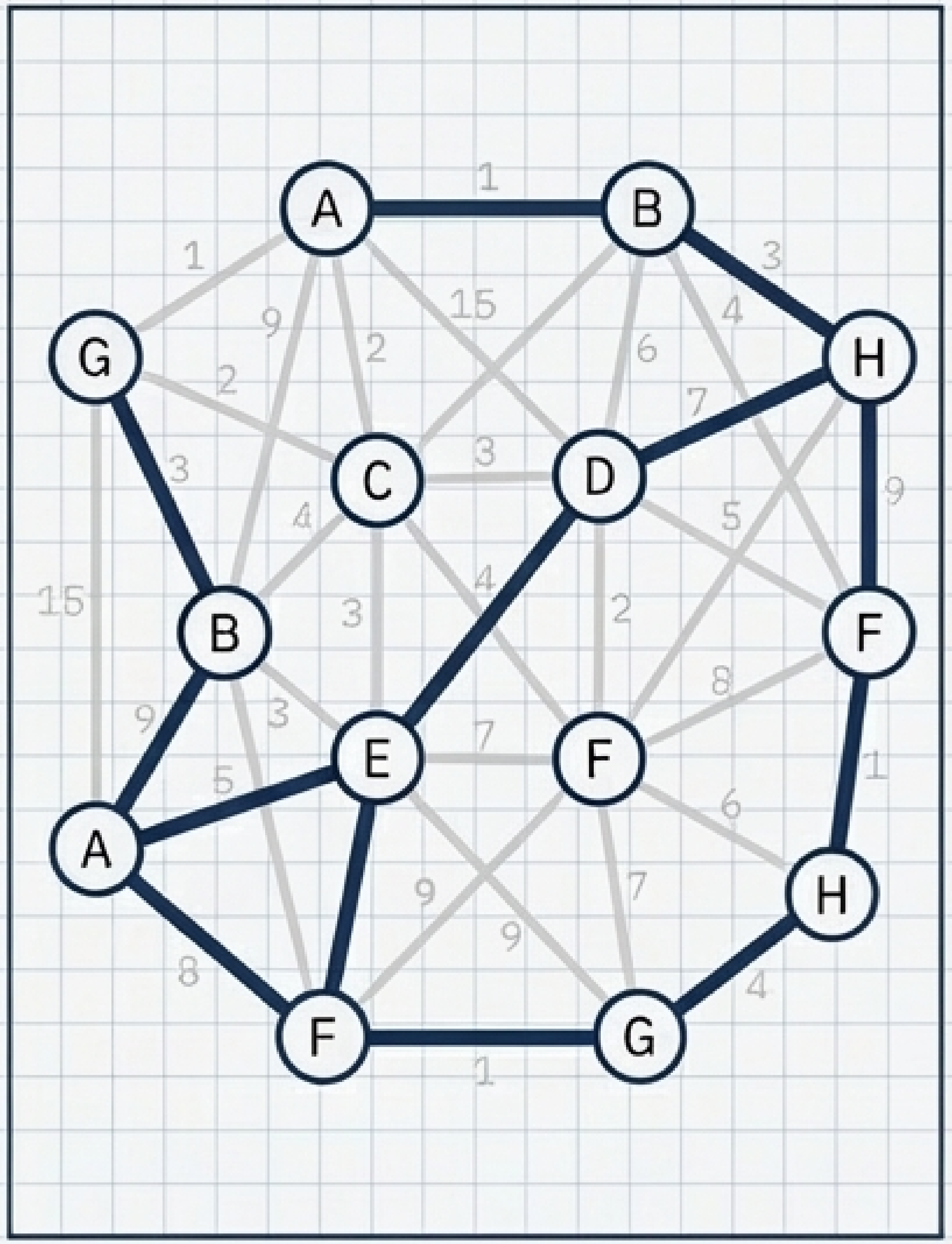
= Unexplored
= Evaluating Neighbors
= Absorbed into MST



Start at arbitrary node.
Evaluate immediate neighbors.



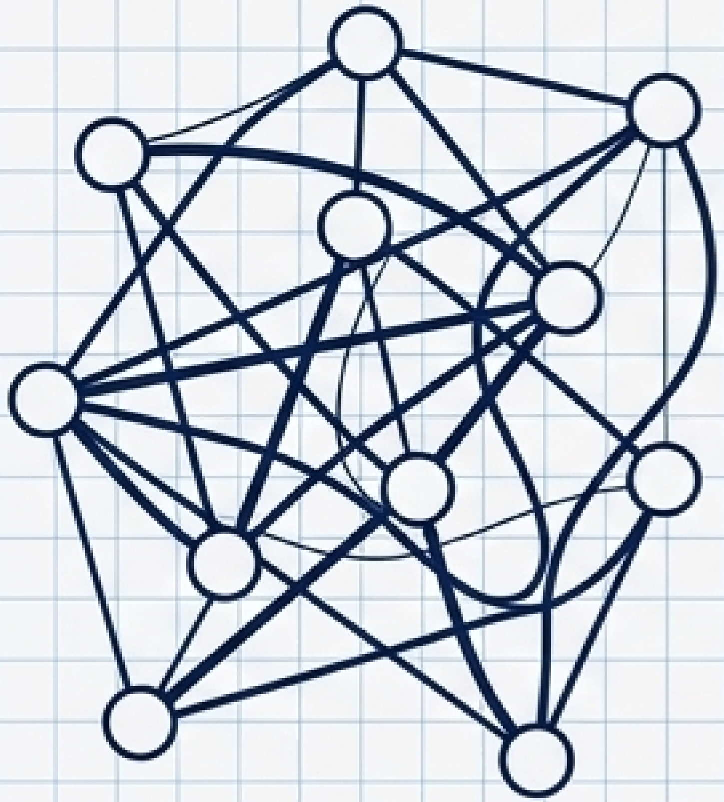
Absorb the closest vertex.
Expand the search horizon.



Stop when all vertices are absorbed.

Helvetica Neue

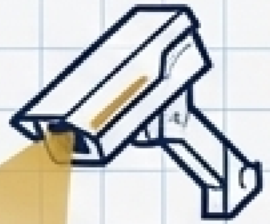
The Graph



Phase 1: Chaos

Translation

The Matrix

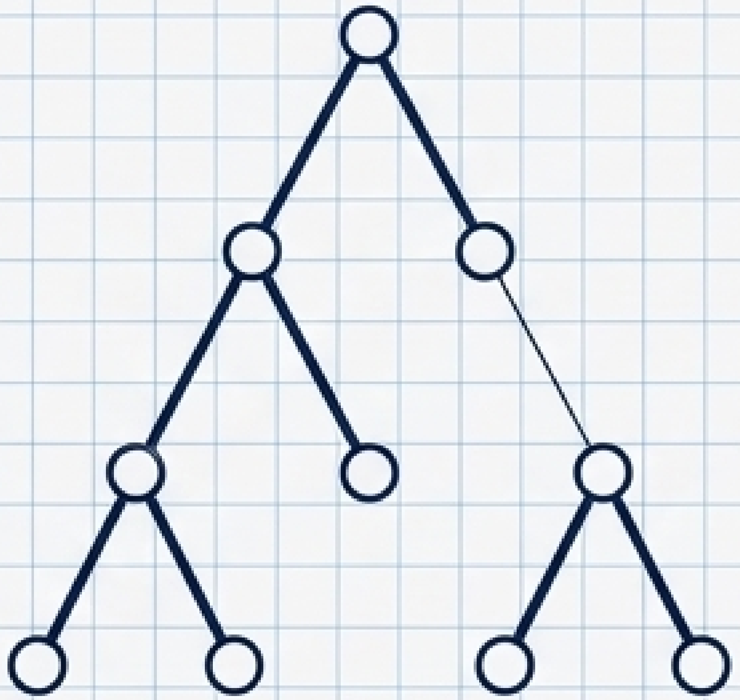


	A	B	C	D	E	F	G	...	I	C	...	V1	V2	V2	V...
A	1	1	1	1	1	1	1	1	0	0	1	0	1	0	1
B	1	1	1	1	0	1	1	1	0	1	1	0	1	0	0
C	0	1	1	1	1	0	1	1	0	1	0	1	0	1	1
D	1	1	1	1	1	1	1	1	1	0	0	0	3	0	1
E	1	1	1	1	1	1	0	1	1	0	1	0	0	1	0
F	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0
...	1	1	1	1	1	1	1	0	1	1	0	1	1	3	1
	1	1	1	1	1	1	0	0	1	1	1	0	0	1	0
...	1	0	1	1	0	1	0	1	0	1	1	1	0	1	0
V1.	1	1	0	0	1	1	0	1	0	0	1	0	1	0	0
V2.	1	0	0	1	0	1	0	0	1	1	0	1	1	0	0
V2...	1	0	1	0	0	1	0	0	1	0	1	0	1	0	1
V2...	1	1	0	0	1	1	0	0	1	3	1	0	0	0	1

Phase 2: Code

Optimization

The Minimal Spanning Tree



Phase 3: Order

The Digital Blueprint: Graph Theory is the framework that allows human architectural logic to be translated into machine-readable matrices, enabling algorithms to automatically extract the hidden, optimized backbone of any network.